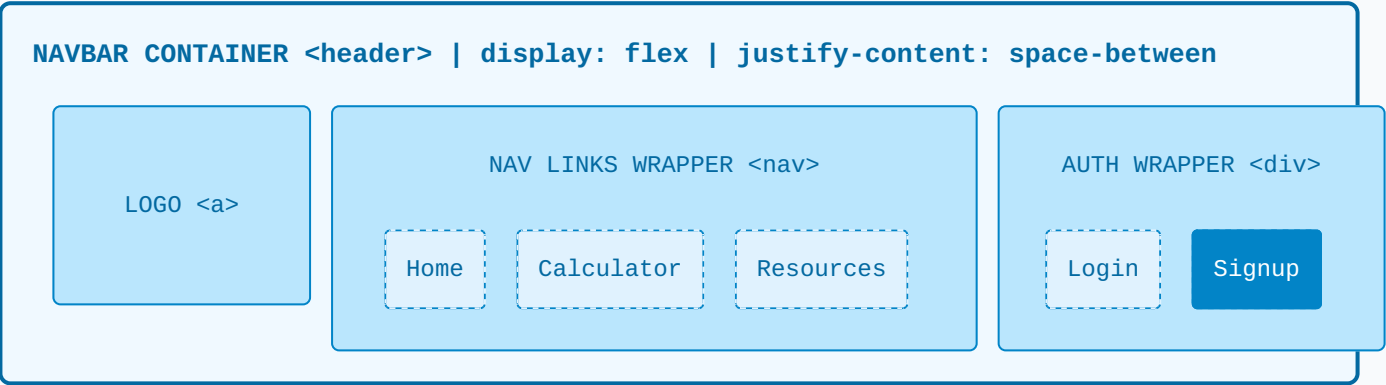


Box Mapping Assignment

Project: Prep-AI Landing Page | Component Hierarchy & CSS Planning

Section 1: Navbar

1. Wireframe Diagram



2. Box Tree - Parent/Child Hierarchy & HTML Tags

Box / Element Name	HTML Tag	Purpose / Notes
Navbar Container	<header>	Outermost semantic header element, wraps all navigation elements.
Logo	<a>	Clickable brand name ("Prep-AI"), links back to homepage.
Nav Links Wrapper	<nav>	Semantic grouping for site navigation items.
Nav Link Items	<a>	Individual page anchors (Home, Calc, Resources).
Auth Wrapper	<div>	Groups the login and signup buttons together in a flex row.

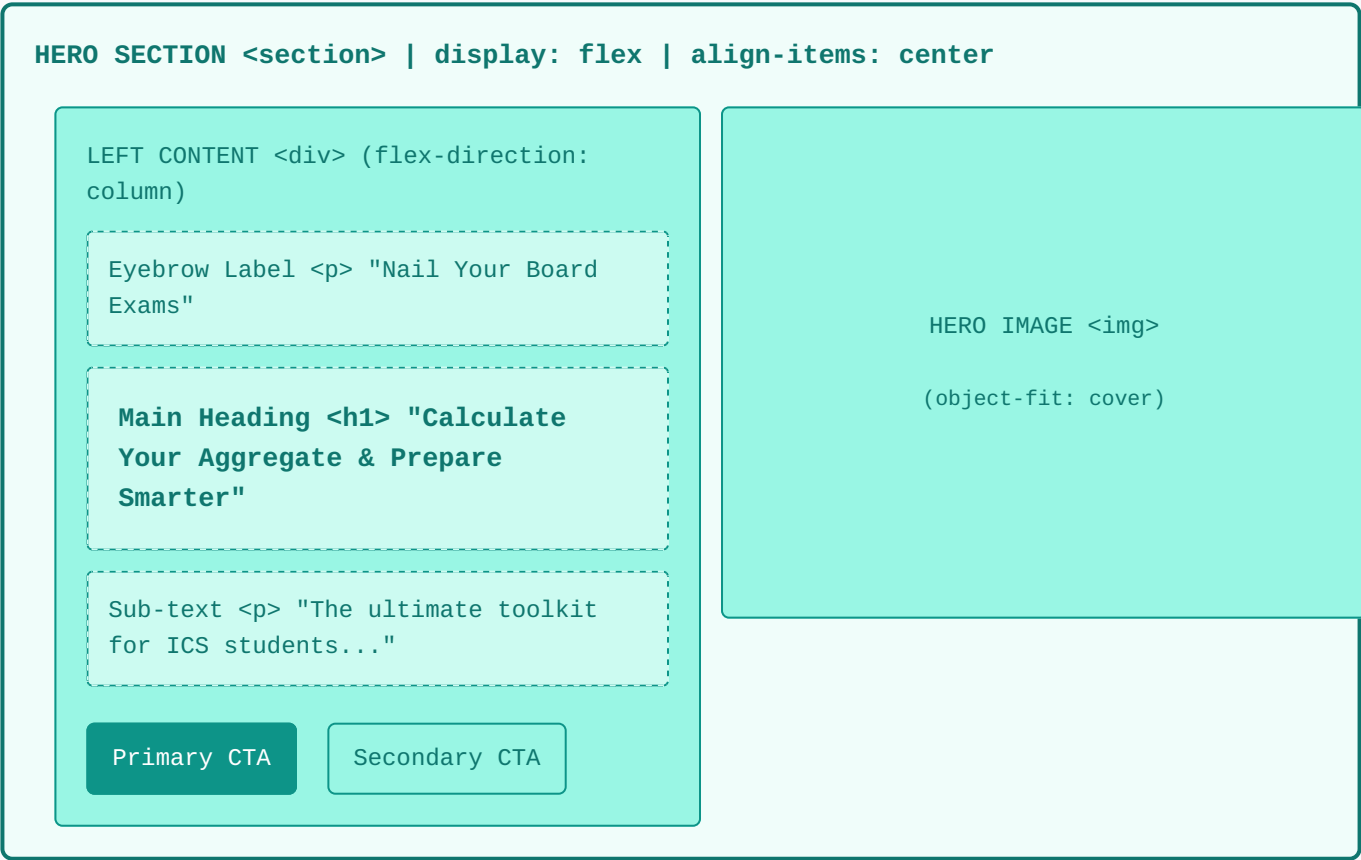
3. CSS Properties & Snippets

Property	Value	Purpose	Code Snippet
display	flex	Puts logo, nav, and auth blocks in one row.	header { display: flex; }
justify-content	space-between	Pushes the 3 main groups to edges and center.	justify-content: space-between;

Property	Value	Purpose	Code Snippet
align-items	center	Vertically aligns text and buttons perfectly.	<code>align-items: center;</code>
position	sticky	Keeps navbar at top when scrolling to the calculator.	<code>position: sticky; top: 0;</code>

Section 2: Hero Section

1. Wireframe Diagram



2. Box Tree - Parent/Child Hierarchy & HTML Tags

Box / Element Name	HTML Tag	Purpose / Notes
HERO SECTION	<code><section></code>	Main wrapper for top fold, id="hero".
LEFT CONTENT	<code><div></code>	Flex column layout for headline and buttons.
Main Heading	<code><h1></code>	Primary page title (Only ONE h1 per page).
CTA WRAPPER	<code><div></code>	Holds both action buttons in a row (display: flex; gap: 16px).

Box / Element Name	HTML Tag	Purpose / Notes
RIGHT IMAGE	<code></code>	Relevant graphic of studying, src and alt required.

3. CSS Properties & Snippets

Property	Value	Purpose	Code Snippet
display	flex	Side-by-side layout for text and image.	<code>section { display: flex; }</code>
flex	1	Left content & right image take 50% width equally.	<code>.left-content { flex: 1; }</code>
flex-direction	column	Stacks the left-side text items vertically.	<code>flex-direction: column;</code>
object-fit	cover	Ensures image fills area without stretching.	<code>img { object-fit: cover; }</code>

Section 3: Aggregate Calculator Tool

1. Wireframe Diagram



2. Box Tree - Parent/Child Hierarchy & HTML Tags

Box / Element Name	HTML Tag	Purpose / Notes
CALCULATOR SECTION	<section>	Wrapper for the calculator, light gray background to pop.
CALC CARD	<div.card>	Elevated white box container serving as the main UI module.
FORM WRAPPER	<form>	Groups inputs for accurate event handling and form submission.
Input Group	<div>	Wraps label and input fields together.

Box / Element Name	HTML Tag	Purpose / Notes
Input Fields	<code><input></code>	Type="number", captures academic marks for calculation.

3. CSS Properties & Snippets

Property	Value	Purpose	Code Snippet
flex-direction	column	Stacks the section header above the calculator card.	<code>section { flex-direction: column; }</code>
align-items	center	Centers the card perfectly in middle of screen.	<code>align-items: center;</code>
box-shadow	0 8px 32px ...	Adds visual depth, lifting the card off the background.	<code>box-shadow: 0 8px 32px rgba(0,0,0,0.1);</code>
max-width	480px	Prevents card inputs from stretching too wide on PC.	<code>.card { max-width: 480px; }</code>
box-sizing	border-box	Ensures padding doesn't widen elements past constraints.	<code>* { box-sizing: border-box; }</code>